

Position-Dependent Curvature for Hyperbolic Embeddings of Medical Ontologies

A Research Proposal

Abstract

Modern medical large language models represent disease concepts in flat Euclidean embedding spaces, despite the established result that tree-structured data cannot be embedded in Euclidean space without distortion that grows with the size of the tree. Recent work has applied constant-curvature hyperbolic embeddings to medical ontologies (Beam et al., 2018; Yang et al., 2025; Kim et al., 2026), demonstrating substantial improvements over Euclidean baselines. However, these methods impose a single global curvature on heterogeneous ontologies whose local branching structure varies substantially across regions — the cardiovascular chapter of ICD-10 has different structure than the external-causes chapter, which has different structure again from the genetic-diseases branch. We propose to derive and evaluate a position-dependent curvature $\kappa(x)$ for hyperbolic embeddings of medical ontologies, framing the problem as embedding into a Cartan-Hadamard manifold of variable negative curvature rather than constant-curvature hyperbolic space. We hypothesize this method will (1) reduce embedding distortion on heterogeneous medical hierarchies, (2) enable lower-dimensional embeddings suitable for resource-constrained clinical deployment, and (3) provide a geometric criterion for hallucination detection in medical LLMs. This proposal outlines the mathematical framework, baseline replication strategy, and downstream evaluation plan; experimental results are forthcoming.

1. Introduction and Motivation

Rare-disease misdiagnosis is a documented clinical crisis. The average rare-disease patient sees roughly seven specialists over five or more years before receiving a correct diagnosis, with cascading effects on treatment outcomes and survival. A significant component of this failure mode is informational: clinicians rely on lookup tools and AI-augmented decision support that struggle to navigate the implicit hierarchy of disease classifications. When a clinician searches for a candidate diagnosis, current systems poorly surface adjacent or cousin diagnoses — diseases that branch off one or two steps away in the medical taxonomy — leading to systematic misses for diseases adjacent to the search target.

Modern medical large language models (LLMs) inherit and amplify this limitation. These models represent disease concepts as dense vectors in flat Euclidean embedding spaces. However, medical ontologies — including ICD-10, SNOMED-CT, and Mondo — are fundamentally hierarchical, and a classical theorem in metric embedding establishes that trees cannot be embedded in Euclidean space without distortion that grows logarithmically in the size of the tree (Linial et al., 1995). Two diseases that are siblings in the medical taxonomy must therefore land either too close, causing confusion, or too far apart, failing to capture relatedness, when embedded in flat space. This is not a bug fixable by training; it is a geometric property of the embedding space.

Hyperbolic geometry, with its exponential volume growth, is the natural setting for tree-structured data. Sarkar (2011) showed that any tree admits a $(1+\epsilon)$ -distortion embedding into the Poincaré disk in linear time. Nickel and Kiela (2017) demonstrated that hyperbolic embeddings of WordNet substantially outperform Euclidean embeddings of much higher dimension; their five-dimensional Poincaré embeddings achieve mean rank approximately five and mean average precision around 0.82 on noun-hierarchy reconstruction, while two-hundred-dimensional Euclidean embeddings achieve mean rank above one thousand and MAP around 0.17. Beam et al. (2018) ported this technique to the ICD-9 hierarchy on a sixty-three-million-member insurance claims dataset, demonstrating that two-dimensional Poincaré embeddings perform comparably to one-hundred-dimensional Euclidean embeddings on the task of reconstructing chapter structure.

Recent work has continued to extend hyperbolic embedding methods to additional medical contexts. Yang et al. (2025) propose HiT and OnT for SNOMED-CT concept retrieval, demonstrating that hyperbolic distance and entailment cones outperform sentence-transformer baselines on out-of-vocabulary clinical queries. Kim et al. (2026), in the GRAIL system, combine hyperbolic patient-trajectory representations with LLM-based reranking for clinical reasoning under hallucination constraints. Across this body of work, however, the curvature parameter is global — typically $K = -1$ — and treated as a hyperparameter rather than as a quantity derived from the structural properties of the data.

Medical ontologies are not uniform. Branching factor in ICD-10 varies from two to over one hundred across nodes, with systematic differences across chapters. Effective tree depth varies similarly. The volumetric requirement at a node of branching factor b is, to leading order, exponential in b , implying that the optimal local curvature should depend on local structure. Forcing all regions of a heterogeneous ontology into a single global curvature is geometrically suboptimal, and the resulting distortion concentrates precisely in the regions where the ontology is least uniform — often the same regions where rare diseases sit. This proposal develops the case that medical ontology embeddings should use a position-dependent curvature $\kappa(x)$ where the local sectional curvature varies smoothly across the embedding manifold to match the local branching structure of the underlying ontology.

2. Background

2.1 Hyperbolic Geometry and Tree Embeddings

Hyperbolic space is the unique simply connected complete Riemannian manifold of constant negative sectional curvature. In the Poincaré ball model, the manifold is the open unit ball in n -dimensional Euclidean space equipped with the conformally rescaled metric:

$$g_x = (4 / (1 - ||x||^2)^2) \cdot g^E$$

where g^E denotes the Euclidean metric. The induced distance between two points u, v in the ball is:

$$d(u, v) = \text{arcosh}(1 + 2||u - v||^2 / [(1 - ||u||^2)(1 - ||v||^2)])$$

Volume in n -dimensional hyperbolic space grows exponentially with radius — $V(r)$ scales as $\exp((n - 1)r)$ — in contrast to polynomial growth in Euclidean space. This volumetric exponential growth is what enables low-distortion tree embeddings: the number of nodes in a tree at depth r grows exponentially in r , matching the available volume in hyperbolic space. Sarkar's construction provides an explicit linear-time algorithm for embedding any weighted tree into the Poincaré disk with worst-case distortion at most $1+\epsilon$ for appropriately scaled edges, recursively placing nodes via Möbius reflections and exploiting the exponential volume growth to avoid sibling collisions even in two dimensions.

2.2 Hyperbolic Embedding Learning

Nickel and Kiela (2017) introduced a method for learning Poincaré embeddings via Riemannian stochastic gradient descent. Given a set D of observed parent-child or related pairs, they minimize a softmax-style negative-sampling loss:

$$L(\Theta) = \sum_{(u,v) \in D} \log [\exp(-d(u, v)) / \sum_{v' \in N(u)} \exp(-d(u, v'))]$$

where $N(u)$ is a set of negative samples drawn uniformly from the vocabulary. Riemannian gradient descent in the Poincaré ball reduces to a rescaling of the Euclidean gradient by the inverse metric tensor. Since the metric is a scalar matrix, the inverse is trivial; the resulting parameter update is:

$$\theta_{t+1} \leftarrow \text{proj}(\theta_t - \eta_t \cdot ((1 - \|\theta_t\|^2)^2 / 4) \cdot \nabla^E L)$$

with the projection operator ensuring embeddings remain inside the open unit ball. The $(1 - \|\theta\|^2)^2 / 4$ multiplier on the Euclidean gradient encodes the curvature of the embedding space.

2.3 Hyperbolic Embeddings in Medicine

Beam et al. (2018) applied Poincaré embeddings to the ICD-9 hierarchy. Their two primary findings: two-dimensional Poincaré embeddings recover the ICD-9 chapter, sub-chapter, and major-code hierarchy with in-category to out-of-category distance ratios comparable to one-hundred-dimensional Euclidean embeddings; and when separate Poincaré embeddings are trained on cohorts with and without metabolic disorders, the relative positions of certain diagnoses (notably pain-related codes around type-2 diabetes) shift in ways consistent with known clinical comorbidity literature.

More recent work has extended hyperbolic embeddings to additional medical settings. Yang et al. (2025) propose hyperbolic transformer-based methods for SNOMED-CT, demonstrating substantial gains over Euclidean baselines for hierarchical retrieval with out-of-vocabulary queries. Kim et al. (2026), in the GRAIL system, propose using hyperbolic patient-trajectory representations as a structured prior to constrain LLM reasoning. In phylogenetic embedding, where similar geometric setups arise, Macaulay et al. (2023) showed empirically that improvement in embedding fidelity saturates around a curvature of $K = -1$ across a wide range, suggesting that for relatively uniform tree structures the choice of curvature is forgiving. Medical ontologies are not uniform, which is the gap this proposal addresses.

3. Problem Statement

We formalize the problem as follows. Let $T = (V, E)$ be a medical ontology with rooted tree (or near-tree) structure, equipped with branching statistics $b: V \rightarrow \mathbb{N}$. We seek an embedding $f: V \rightarrow M$ into a Riemannian manifold (M, g) such that:

(a) the graph distance d_T on the tree is approximated by the Riemannian distance d_g on M with low multiplicative distortion;

(b) the manifold (M, g) is a Cartan-Hadamard manifold (complete, simply connected, non-positive sectional curvature) so that geodesics are unique and exponential maps are global diffeomorphisms;

(c) the sectional curvature $\kappa(x)$ varies smoothly across M to match the local branching structure of T at the corresponding embedded points;

(d) the Riemannian optimization required to learn the embedding remains computationally tractable on commodity hardware.

Existing methods correspond to the special case $\kappa(x) \equiv K_0 < 0$ constant. The novel content of this proposal is the construction, derivation, and empirical evaluation of a non-trivial $\kappa(x)$ for medical ontologies.

4. Proposed Approach

4.1 Mathematical Framework

We propose to work in a conformally Euclidean variable-curvature manifold. Consider M as an open subset of $\mathbb{R}^n$ equipped with the metric:

$$g_x = (1 / \lambda(x)^2) g^E$$

where $\lambda(x)$ is a smooth positive function generalizing the conformal factor $(1 - ||x||^2)/2$ of the Poincaré ball. The sectional curvature of this metric is determined by λ via a quasi-linear elliptic operator; for the conformally flat case the formula reduces, up to dimensional factors, to a combination of $|\nabla \log \lambda|^2$ and $\Delta \log \lambda$. By choosing $\lambda(x)$ appropriately, we can prescribe a target curvature $\kappa(x)$ within wide limits; the inverse problem of recovering λ from a desired κ is a well-posed PDE that can be solved analytically in special cases or numerically otherwise.

Our specific proposal: derive $\kappa(x)$ from the local branching factor of the ontology. Concretely, if a region of the ontology has local branching factor b , the volumetric requirement is V proportional to b^d at depth d , which in a Cartan-Hadamard manifold of curvature K corresponds to volume growth at rate $\exp(r \cdot \sqrt{|K|})$. Equating these gives a first-order target $K(b)$ of order $-(\log b)^2$. This first-order target requires refinement: the actual derivation must account for the dimensional dependence of volume growth, smoothness constraints on $\kappa(x)$, and consistency conditions where neighboring regions have different branching factors. The full derivation is a planned deliverable.

4.2 Embedding Optimization

The Riemannian gradient update generalizes from the constant-curvature case:

$$\theta_{t+1} \leftarrow R_{\theta_t}(-\eta_t \cdot g^{-1}(\theta_t) \nabla^E L)$$

where $g^{-1}(\theta_t)$ now varies with position and R denotes a retraction onto M . We propose using either the exponential map $\exp_\theta$ when computationally feasible (typically requiring numerical integration of the geodesic equation) or a projection-based retraction otherwise. The negative-sampling loss from Nickel and Kiela carries over directly, with the distance $d(u, v)$ computed via geodesic-equation integration in the variable-curvature metric. We will build on the `geoopt` library (Bécigneul and Ganea, 2019) for the Riemannian optimization infrastructure.

4.3 Distortion Analysis

A central theoretical contribution will be a distortion bound for variable-curvature embeddings of trees with non-uniform branching. The hypothesis: for an ontology with maximum branching factor $b_{\max}$ and a position-dependent curvature satisfying $\kappa(x) \leq -(\log b_{\text{local}})^2$ in the appropriate sense, the distortion of a Sarkar-style construction is improved over the constant-curvature case for non-uniform branching, with the magnitude of the improvement growing with the branching variance. Formalization of this bound is a planned deliverable.

4.4 Hallucination Detection

A secondary hypothesis: hallucinated medical codes generated by an LLM, when embedded in the variable-curvature space using the same procedure as legitimate codes, should land geometrically far from the entailment cones of the patient's actual condition. This provides a geometric criterion for hallucination detection without requiring labeled hallucination data for training. We propose to test this on the Med-HALT benchmark and on adversarial code-generation tasks evaluated against MIMIC-IV ground truth, comparing against perplexity-based and self-consistency-based baselines.

5. Methodology and Experimental Plan

5.1 Phase 1: Baseline Replication

The first concrete deliverable is faithful replication of Beam et al.'s ICD-9 reconstruction results, ported to ICD-10-CM. This requires parsing the ICD-10-CM tabular format into a parent-child edge list (approximately 70,000 nodes and 70,000 edges across multiple hierarchical levels); training Poincaré embeddings via `geoopt` or the original Nickel-Kiela implementation; and reproducing reconstruction metrics (mean rank, MAP) and the in-category versus out-of-category distance comparison. The target is performance comparable to the published ICD-9 results, mirroring Beam et al.'s observation that two-dimensional Poincaré embeddings approach one-hundred-dimensional Euclidean performance on hierarchy reconstruction. This phase establishes the constant-curvature baseline against which all subsequent improvements are measured.

5.2 Phase 2: Variable-Curvature Derivation and Implementation

Derive $\kappa(x)$ from local branching statistics of ICD-10 and implement training in the resulting variable-curvature manifold. Compare against the Phase 1 baseline on the same reconstruction protocol, controlling for embedding dimension, optimizer, and

hyperparameters. The hypothesis: variable-curvature embeddings achieve lower distortion at any given dimension, with the largest improvements concentrated in the most heterogeneously branched chapters of the ontology. Numerical stability near the boundary of the manifold is a known risk; we will address this through gradient clipping and careful initialization, falling back to constant-curvature when necessary to isolate sources of instability.

5.3 Phase 3: Downstream Evaluation on Rare-Disease Retrieval

Use the Human Phenotype Ontology (HPO) and Mondo Disease Ontology to construct a symptom-to-disease retrieval benchmark on Orphanet rare diseases. Embed both HPO terms and Mondo disease entities in the variable-curvature space. Evaluate retrieval accuracy (top-k recall, mean reciprocal rank) compared to constant-curvature and Euclidean baselines. The intended demonstration: rare diseases whose phenotypic presentation overlaps with more common diseases are poorly retrieved by Euclidean and (to a lesser extent) constant-curvature methods, but are retrievable when the embedding geometry respects the local branching structure of the rare-disease subtree.

5.4 Phase 4: Hallucination Detection on Med-HALT

Embed model-generated medical codes in the trained space; compute geometric distance to the patient's contextual entailment cone; evaluate as a hallucination signal on Med-HALT, with comparison to perplexity-based and self-consistency baselines. The metric of interest is detection AUROC at fixed false-positive rate.

5.5 Datasets

Open and instantly available: ICD-10-CM (CMS, public domain), Mondo Disease Ontology (CC-BY 4.0), Human Phenotype Ontology and HPOA annotations, Orphanet rare-disease ontology, MedQA, and Med-HALT. Free with registration: UMLS Metathesaurus. Credentialed access required: MIMIC-IV (PhysioNet), used in Phase 4 for ground-truth validation. SNOMED-CT is optional and used only as a stretch-goal alternative ontology.

5.6 Compute Requirements

All training is feasible on a single GPU. Phase 1 baseline replication requires only modest compute (T4-class GPU sufficient for embedding dimension up to 100). Phase 2 variable-curvature training is more expensive due to numerical geodesic integration; an A100 GPU on Google Colab Pro+ is sufficient for embedding dimensions up to 50, which is well within the relevant range for ontology embedding tasks.

6. Expected Contributions

If successful, this work will produce: (C1) an explicit derivation of a position-dependent curvature function $\kappa(x)$ for hierarchical embeddings of medical ontologies, motivated by local branching statistics; (C2) an implemented and benchmarked variable-curvature Riemannian optimization framework for ontology embedding, building on geoopt; (C3) empirical demonstration of improved embedding quality (reduced distortion, lower required dimension) on ICD-10 and Mondo over constant-curvature baselines; (C4) a

geometric hallucination-detection method for medical LLMs, evaluated on Med-HALT; and (C5) open-source code and trained embeddings released under permissive license to support reproducibility.

7. Limitations and Risks

The principal mathematical risk is that the position-dependent metric defined by $\lambda(x)$ may not satisfy the Cartan-Hadamard conditions globally, requiring modifications to ensure non-positive curvature everywhere. The principal computational risk is numerical instability in geodesic integration near regions of rapidly varying curvature, which we address through adaptive integration step size and gradient clipping. The principal experimental risk is that improvements over constant-curvature baselines may not generalize across all chapters or all embedding dimensions; we address this by reporting per-chapter results and by structuring experiments to isolate where variable curvature does and does not help. Throughout, the constant-curvature baseline is maintained as the fallback comparison, ensuring that even negative results contribute information about the structure of medical ontologies.

8. Timeline

Phase 1 (baseline replication): months 1-2. Phase 2 (variable-curvature derivation and implementation): months 3-5. Phase 3 (rare-disease retrieval): months 5-7. Phase 4 (hallucination detection): months 7-8. Writing and revision: months 8-9. Submission to Regeneron Science Talent Search and follow-on venues: month 10.

9. Conclusion

Hyperbolic embeddings have established themselves as the appropriate geometry for hierarchical medical data, with strong empirical results across ICD, SNOMED, and patient-trajectory representations. However, existing methods uniformly assume constant curvature, ignoring the substantial heterogeneity of real medical ontologies. We propose to derive, implement, and empirically validate a position-dependent curvature method that respects this heterogeneity, and to demonstrate its utility for both ontology embedding quality and downstream LLM applications, with particular attention to the high-value clinical problem of rare-disease retrieval and medical-LLM hallucination detection.

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